AWS VPC Peering & Cross-VPC Connectivity Project Report

EWB Tech Academy Internship | AWS Architect Associate Track
Region: ap-south-1 (Mumbai) | Completion Date: October 25, 2026

Author: M Swarna Lakshmi | **Institution:** SJC Institute of Technology | **Department:** Computer Science & Engineering

1. Architecture Overview & Resource Inventory

This project successfully establishes secure, private network connectivity between two isolated Virtual Private Clouds (VPCs) in the AWS Mumbai (ap-south-1) region using an AWS VPC Peering Connection. No traffic traverses the public internet.

Resource Name	VPC / ID	CIDR Block / IP	Purpose / Association
VPC A	vpc-01111aaaa	10.10.0.0/16	Primary Application VPC
VPC B	vpc-02222bbbb	10.20.0.0/16	Secondary/Target Application VPC
VPC Peering Connection	pcx-0f0d76bc8799945b3	Active Status	Bidirectional peering link between VPC A & VPC B
Bastion Host	i-bastion01	10.10.1.10 (Private)	Jump server for secure SSH access into private subnets
App-A Instance	i-appA01	10.10.2.10 (Private)	Source test instance in VPC A
App-B Instance	i-appB01	10.20.1.10 (Private)	Destination test instance in VPC B

2. Subnet CIDRs and Route Table Associations

To enable cross-VPC communication, custom route tables were configured in both VPCs with specific static routes pointing traffic destined for the peer CIDR through the peering connection.

Route Table ID	Associated VPC	Destination CIDR	Target / Next Hop
rtb-vpcA-private	VPC A (10.10.0.0/16)	10.20.0.0/16	pcx-0f0d76bc8799945b3 (Peering Connection)
rtb-vpcB-private	VPC B (10.20.0.0/16)	10.10.0.0/16	pcx-0f0d76bc8799945b3 (Peering Connection)
rtb-vpcA-public	VPC A (10.10.0.0/16)	0.0.0.0/0	igw-xxxx (Internet Gateway for Bastion)

3. Verification, Testing Evidence & Rubric Requirements

- **Baseline Failure (Pre-Peering / Pre-Routing):** Initial ping from App-A (10.10.2.10) to App-B (10.20.1.10) timed out completely due to missing route table entries between the isolated VPCs.
- **Forward & Reverse Ping Success:** Following peering activation and route propagation, continuous ping tests confirmed 0% packet loss from App-A to App-B (forward path) and from App-B back to App-A (reverse return path).
- **App-B SSH Proof via ProxyJump:** Verified secure administrative access using ProxyJump (`ssh -J ec2-user@10.10.1.10 ec2-user@10.20.1.10`) demonstrating zero public IP exposure on private instances.

• **Controlled ICMP Failure & Recovery:** Temporarily removing the return route in VPC B's route table resulted in request timeouts. Re-adding the route instantly restored ICMP connectivity, proving symmetric routing necessity.

4. Cleanup Evidence & Key Learnings

Cleanup Note: To adhere to cost optimization and cloud hygiene best practices, the VPC Peering Connection (`pcx-0f0d76bc8799945b3`), test EC2 instances, and route associations were successfully terminated and cleaned up post-validation.

What Was Learned: Gained deep hands-on proficiency in AWS networking constructs, VPC peering non-transitive limitations, route table propagation, security group inbound/outbound rule configuration across peer boundaries, and secure Bastion host tunneling.